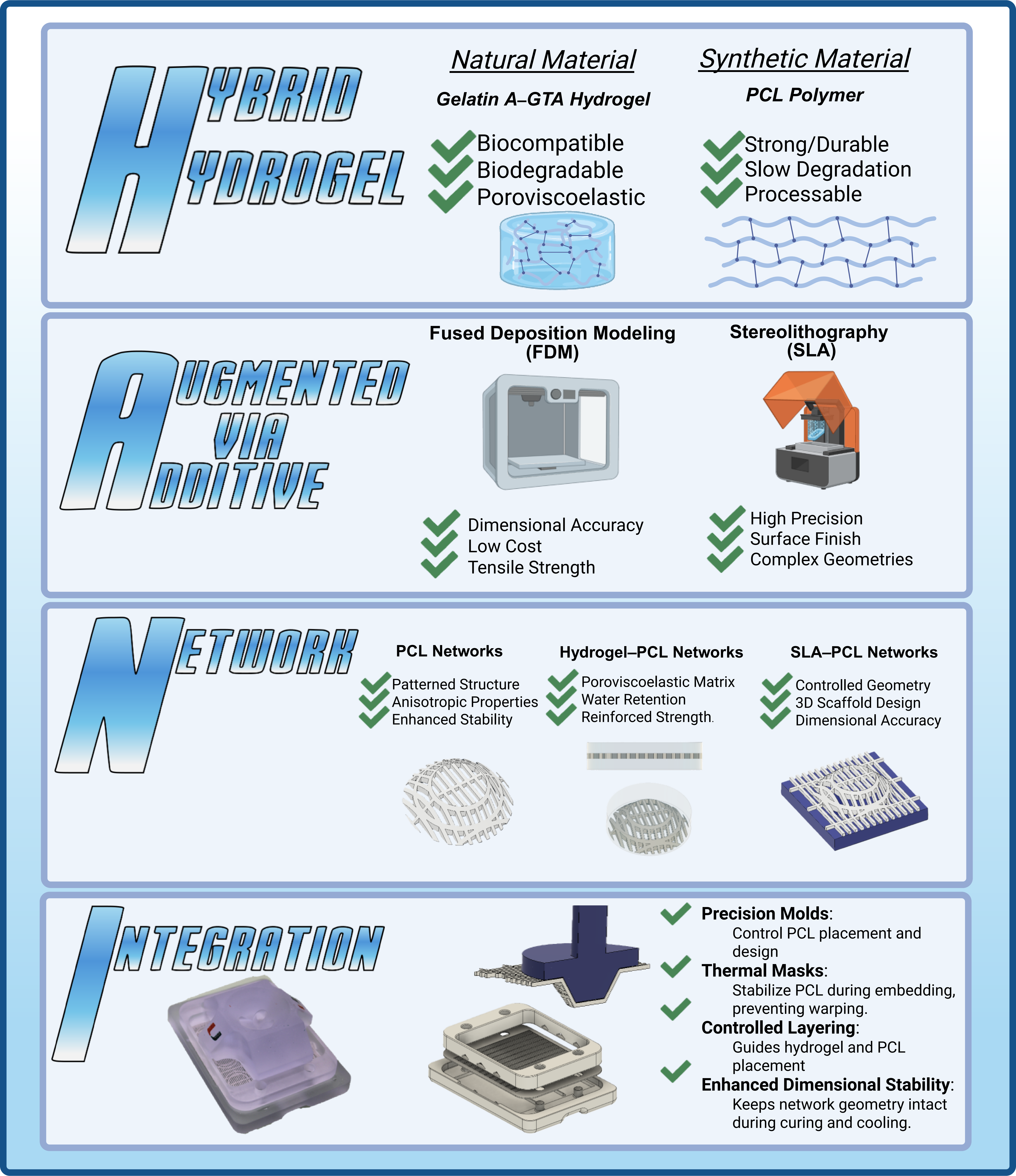
HYBRID HYDROGEL
Natural Material
Gelatin A–GTA Hydrogel
Biocompatible
Biodegradable
Poroviscoelastic
Synthetic Material
PCL Polymer
Strong/Durable
Slow Degradation
Processable
AUGMENTED VIA ADDITIVE
Fused Deposition Modeling (FDM)
Dimensional Accuracy
Low Cost
Tensile Strength
Stereolithography (SLA)
High Precision
Surface Finish
Complex Geometries
NETWORK
PCL Networks
Patterned Structure
Anisotropic Properties
Enhanced Stability
Hydrogel–PCL Networks
Poroviscoelastic Matrix
Water Retention
Reinforced Strength.
SLA–PCL Networks
Controlled Geometry
3D Scaffold Design
Dimensional Accuracy
INTEGRATION
Precision Molds: Control PCL placement and design
Thermal Masks: Stabilize PCL during embedding, preventing warping.
Controlled Layering: Guides hydrogel and PCL placement
Enhanced Dimensional Stability: Keeps network geometry intact during curing and cooling.